

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Product Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built data-backed automation flows that combined generated drafts, public-site API lookups, validation, retries, and operator review before execution.
- Reduced manual filing time by **85%** through backend state checks, failure recovery, and repeatable browser automation.
- Deployed Dockerized services through Google Cloud Run and GitHub Actions, improving iteration speed by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python ML tooling for MRI segmentation across **95 anatomical regions**, covering preprocessing, inference integration, generated masks, and large-array data paths.
- Reduced runtime by **87%** using asynchronous and parallel execution over expensive NumPy/PyTorch-style workflows.
- Validated model outputs through inspectable masks and 3D surfaces while debugging label mapping and data-orientation issues.

Open Source

Contributions 

- Contributed to **Kuadrant MCP Gateway** backend-outage grace handling, **Kyverno** policy/infrastructure paths, and IOOS/Tiled data or visualization tooling under maintainer review.

Projects

Hyoka | *Python, FastAPI, PostgreSQL, OpenTelemetry, SQLAlchemy, Typer, Docker*

- Built an **AI-agent reliability control plane** with SDK/proxy/OTLP trace ingestion, evaluator registries, replay modes, failure mining, validation runs, release gates, promotions, and audit logs.
- Implemented worker-owned leases, content-addressed artifacts, signed manifests, project API keys, SQLAlchemy/Alembic models, and reproducible runs around immutable inputs.

Swish Support System | *Python, FastAPI, React, PostgreSQL/pgvector, Redis, Langfuse*

GitHub 

- Built a hybrid AI + rules support workflow where LLM assessments classify ambiguous complaints while deterministic policy code controls refunds, replacements, coupons, and escalations.
- Grounded responses in order, item, evidence, customer, and policy context with pgvector retrieval, Redis state, Langfuse traces, confidence thresholds, and regression cases.

Penny Lane Capital | *Python, Agno, Pydantic, SQLite, Vector Memory, yfinance, JSONL*

GitHub 

- Built a Python multi-agent research workflow with analyst roles, bull/bear debate, trader, risk review, portfolio approval, typed state, checkpoints, and replayable JSONL traces.
- Evaluated decisions with historical-market backtests, reward-loop scorecards, semantic lesson retrieval, and return, drawdown, win-rate, and risk signals.

Technical Skills

Languages: Python, TypeScript, Go, SQL, JavaScript

Frameworks: FastAPI, React, Next.js, Node.js, Agno, PyTorch, ONNX Runtime

Tools: PostgreSQL/pgvector, Redis, SQLite, Vector Memory, Docker, GCP, GitHub Actions, OpenTelemetry, Langfuse, JSONL